

FinTech Capstone

Executive Summary & Dashboard Analysis Report

Debt Burden, Cashflow Buffer, Credit Risk, Loan Management & Informal Business Finances — a five-cluster analysis of FinTech.csv, with a recommended MVP direction

Prepared for: Group 3 Capstone — Product & Data Analytics Team

Sample size: 32,424 users | Source: FinTech.csv

Executive Summary

This report accompanies the interactive Power BI dashboard built against FinTech.csv (32,424 users) and explains every visual on it in plain language, alongside the insight, conclusion, and recommendation each one supports. The dashboard investigates five candidate financial problems — debt burden, cashflow buffer strain, credit risk, loan management, and informal business finances — using only what the dataset can directly verify.

The analysis surfaced one dominant, cross-cutting finding: every risk factor examined is broadly distributed across the population rather than concentrated in any single region, age group, gender, education level, or job title. This held true independently across all five clusters, which strengthens rather than weakens its credibility — five separate analyses converging on the same structural conclusion is a strong signal, not a coincidence.

Based on population size affected, severity, and — critically — how actionable each pattern is as a product feature, this report recommends three MVP candidates for the capstone build, ranked and justified in the final section: (1) Debt Burden for Employed Borrowers, (2) Thin Cashflow Buffer, and (3) Credit-Invisible / Subprime Population.

Reading note: every dashboard number in this report has been checked against the raw FinTech.csv sample directly. Where a real-world Nigeria population estimate is referenced for context, it is clearly marked as external, since FinTech.csv cannot itself verify a national figure.

Methodology & Data Quality Notes

FinTech.csv contains 32,424 synthetic user records with fields covering income, expenses, savings, loan details (type, amount, term, interest rate, EMI), credit score, employment status, job title, education level, gender, age, and region. Every metric in this report is computed directly from these fields — no external data was blended into the dashboard's KPI values themselves.

One limitation was identified early and respected throughout: the dataset's fields behave largely independently of one another (for example, credit score shows almost no relationship with the interest rate charged). This means the dashboard reports what percentage of users fall into a given risk band with full confidence, but does not claim causal relationships the data cannot support — for example, it does not claim that a specific demographic causes a specific risk, only that no demographic differentiates exposure to it.

Cluster A — Debt Burden for Employed Borrowers

Headline finding: 82.2% of employed users with an active loan carry a debt-to-income (DTI) ratio above 0.4 — a widely used financial-risk threshold.

Visual 1 — KPI Card: Pct High DTI

What it shows: A single headline number: 82.2% of employed loan-holders exceed the 0.4 DTI threshold.

What it means: This is the sharpest, single most alarming number in the entire analysis. More than 4 in 5 employed people carrying a loan are financially over-extended by a standard industry benchmark — this is not a marginal or edge-case problem.

Visual 2 — Severity Bar Chart: Count of user_id by DTI_Band

What it shows: Employed loan-holders broken into four severity bands: Healthy (under 0.4), Elevated (0.4–1.0), High (1.0–3.0), and Severe (3.0+).

What it means: The distribution skews heavily toward High and Severe rather than clustering just over the 0.4 line. This is not a population hovering near a soft threshold — a large share are deep into risk territory, which argues for an intervention that addresses meaningful debt reduction, not just a one-time warning.

Visual 3 — Segmentation Chart: Pct High DTI by Region / Age / Gender / Education / Job Title

What it shows: The same 82.2% rate, broken out five different ways using an interactive field parameter.

What it means: Every segment lands within a few points of 82.2% — no region, age band, gender, education level, or profession stands out as more or less exposed. The problem is structural (tied to having a loan while employed), not demographic. This is an important product decision point: the MVP should target all employed borrowers broadly, not a narrow persona.

Visual 4 — 100% Stacked Bar: DTI_Band by Loan Type

What it shows: DTI severity broken down by loan purpose — Business, Car, Education, Home.

What it means: All four loan types show virtually identical severity splits. Loan purpose does not predict risk, reinforcing that any feature here needs to work across all loan types, not just the largest category.

Visual 5 — 100% Stacked Bar: DTI_Band by Loan Term

What it shows: DTI severity broken down by loan term length, from 12 to 360 months.

What it means: This is the one variable that meaningfully differentiates severity in this cluster. Short-term loans (12–36 months) skew heavily toward High/Severe DTI, while long-term loans (120+ months) skew toward Healthy/Elevated — an intuitive, explainable pattern, since shorter terms mechanically require higher monthly payments for the same loan amount. This is the most actionable single insight in the dashboard: term length, not who is borrowing, is the lever.

Cluster A conclusion: This is a large, severe, and broadly distributed problem with one clear, explainable driver (loan term length) that a feature can act on directly — for example, flagging short-term loans as high-risk at the point of origination, or nudging users toward refinancing into longer terms.

Cluster D — Thin Cashflow Buffer / Spending Squeeze

Headline finding: 33.3% of all users spend over 70% of their monthly income; 8.5% spend over 85%, leaving almost no cushion for an unexpected expense.

Visual 1 — KPI Card: Pct Thin Buffer

What it shows: 33.3% of the full user population (not a subgroup) is spending more than 70% of monthly income.

What it means: Unlike Cluster A, this problem applies to everyone in the dataset, not just employed loan-holders — it is the broadest-reaching cluster by population base.

Visual 2 — Severity Bar Chart: Count of user_id by Buffer_Band

What it shows: Users split into four bands: Comfortable (under 50%), Manageable (50–70%), Tight (70–85%), Squeezed (85%+).

What it means: This is a gradient, not a cliff-edge — users slide through Comfortable to Manageable to Tight to Squeezed rather than suddenly becoming at-risk. Over 1,300 users already sit in "Squeezed." A gradient shape is good news for product design: it means an early-warning feature has room to intervene before someone reaches the worst band, rather than only being useful after the fact.

Visual 3 — Segmentation Chart: Pct Thin Buffer by Region / Age / Gender / Education / Job Title

What it shows: The 33.3% rate broken out across the same five demographic cuts.

What it means: Same conclusion as Cluster A: every segment sits within a few points of 33.3%. No profession, region, age band, gender, or education level is disproportionately squeezed. This is a spending-behavior problem, not a segment-specific one.

Visual 4 — 100% Stacked Bar: Buffer_Band by Employment Status

What it shows: Buffer severity broken down by Employed, Self-employed, Student, and Unemployed.

What it means: All four employment types show nearly identical distributions. This directly answers the depth question posed at the start of this analysis — self-employed users are not more likely to have thin buffers than salaried employees. Employment stability does not predict this risk.

Cluster D conclusion: This is the broadest-reaching problem by population, has a natural early-warning structure (a gradient rather than a threshold), and maps directly to the capstone brief's explicit call-out for "spending alerts" as a feature category.

Cluster B — Credit-Invisible / Subprime Population

Headline finding: 50.7% of all users have a credit score below 580; 36.2% fall below 500 — both widely used subprime cutoffs.

Visual 1 — KPI Card: Pct Subprime

What it shows: 50.7% of the entire user base — the largest single population of any cluster in this analysis.

What it means: This is more than half the platform's users. If this were a live product, credit accessibility would be a majority-population concern, not a niche one.

Visual 2 — Severity Bar Chart: Count of user_id by Credit_Band

What it shows: Five bands: Poor (under 500), Subprime (500–580), Fair (580–670), Good (670–740), Excellent (740+).

What it means: The distribution is not evenly spread across risk tiers — "Poor" alone accounts for over 10,000 users, dwarfing every other band. Risk here is concentrated at the most severe end of the spectrum, not mildly distributed across the subprime range.

Visual 3 — Segmentation Chart: Pct Subprime by Region / Age / Gender / Education / Job Title

What it shows: The 50.7% rate broken out across the same five demographic cuts.

What it means: Consistent with every other cluster: no segment differentiates exposure. Subprime credit risk is structural and population-wide, not concentrated in any identifiable group.

Cluster B conclusion: The largest affected population of any cluster, with a severity distribution skewed toward the worst tier. However, this is fundamentally an informational/educational problem (users need to understand and improve their score) rather than a behavioral-nudge problem like Clusters A and D — this affects its MVP feasibility, discussed in the recommendation below.

Cluster C — Debt and Loan Management

Headline finding: 40.1% of all users hold an active loan of some kind.

Visual 1 — KPI Card: Pct Has Loan

What it shows: 40.1% of all users currently carry an active loan.

What it means: A large, but not majority, share of the population. This cluster measures loan prevalence, distinct from Cluster A, which measures loan risk severity among a subgroup.

Visual 2 — Composition Bar Chart: Count of user_id by Loan Type

What it shows: Among loan-holders, the split across Home, Education, Business, and Car loans.

What it means: The four loan types are almost perfectly evenly split (24–25% each). No single loan product dominates, meaning a loan-management feature would need to support all four categories rather than optimizing for one.

Visual 3 — Segmentation Chart: Pct Has Loan by Region / Age / Gender / Education / Job Title

What it shows: The 40.1% rate broken out across all five demographic cuts.

What it means: The flattest result of any cluster in this analysis — every one of the five segments clustered tightly around 40.1% with no deviation. This is the fourth of five clusters confirming the same structural, non-demographic pattern.

Cluster C conclusion: This cluster substantially overlaps with Cluster A — both are built on the same has_loan population, just measured differently (prevalence vs. severity). It does not represent an independent product opportunity; its findings support and reinforce Cluster A rather than standing alone.

Cluster E — Small & Informal Business Finances

Headline finding: 20.3% of all users are self-employed.

Visual 1 — KPI Card: Pct Self Employed

What it shows: 20.3% of the full user population is self-employed — the smallest population base of any cluster.

What it means: This is a real, sizeable segment, but the smallest starting point among all five clusters analyzed.

Visual 2 — Composition Bar Chart: Count of user_id by Buffer_Band (filtered to Self-Employed)

What it shows: The same cashflow-buffer bands from Cluster D, filtered to only self-employed users.

What it means: The shape closely mirrors the general population's buffer distribution from Cluster D — self-employed users are not disproportionately squeezed.

Visual 3 — Segmentation Chart: Pct Self Employed by Region / Age / Gender / Education / Job Title

What it shows: The 20.3% rate broken out across all five demographic cuts.

What it means: Consistent with the rest of the analysis: no segment shows a meaningfully different self-employment rate.

Cluster E conclusion: This directly confirms Cluster D's depth-question finding rather than revealing a new, independent pattern — self-employment does not correlate with greater cashflow strain in this dataset, and the population base is the smallest of the five clusters examined.

Cross-Cutting Finding: A Structural, Not Demographic, Problem Set

The single most important finding in this entire analysis is not any one cluster's headline number — it is what happened when all five clusters were segmented the same five ways (region, age, gender, education level, job title). In every case, without exception, the risk factor was spread evenly across the population.

Five independent analyses converging on the identical conclusion is a strong, defensible pattern rather than coincidence. It carries a direct product implication: none of these five problems require narrow persona targeting. A feature addressing any of them should be built for the broad population, not tuned to a specific segment.

MVP Recommendation — Top 3 Candidates, Ranked

The recommendation below weighs three factors for each cluster: the size of the population affected, the severity/actionability of the pattern, and — most decisively — whether the finding translates into a concrete, buildable feature rather than remaining purely informational.

Rank	Candidate	Population Affected	Why It Ranks Here
1	Debt Burden for Employed Borrowers (Cluster A)	82.2% of employed loan-holders (19.6% of full sample)	Sharpest, most severe number in the analysis, and the only cluster with a clear, explainable, actionable lever: loan term length. A feature can flag short-term loans at origination or nudge refinancing toward longer terms — a concrete, buildable intervention, not just an alert.
2	Thin Cashflow Buffer / Spending Squeeze (Cluster D)	33.3% of all users (broadest population base of any cluster)	Largest reach of any cluster, a natural gradient structure ideal for an early-warning feature, and a direct match to the capstone brief's explicit call-out for spending alerts. Confirmed independent of employment type via Cluster E.
3	Credit-Invisible / Subprime Population (Cluster B)	50.7% of all users (largest single population of any cluster)	The single largest affected population, with risk concentrated at the most severe tier. Ranked third rather than first because the natural feature here (credit education / score-improvement guidance) is more informational than behavioral, making it a slower, less differentiated MVP than a real-time alert or intervention feature.

Why Clusters C and E were not selected

Cluster C (Debt and Loan Management, 40.1%) is built on the same underlying loan population as Cluster A and does not reveal an independent pattern beyond what Cluster A already covers in more depth — building both would mean shipping two features off nearly the same data story. Cluster E (Self-Employed Finances, 20.3%) has the smallest population base of the five and its own analysis confirms it is not a distinct risk profile — self-

employed users mirror the general population almost exactly. Both remain valuable supporting context, but neither earns a place in the top 3 MVP slate.

Final Recommendation

Build the MVP around Cluster A (Debt Burden) as the primary feature, given its severity and its uniquely actionable driver. Use Cluster D (Cashflow Buffer) as the natural second phase — its gradient structure and broad population make it a strong complementary "early warning" layer that extends the same core value proposition (helping users avoid financial over-extension) to the full user base, not just borrowers. Cluster B (Credit Risk) is recommended as a third-phase or parallel content/education feature, given its scale, but should not compete for the same initial engineering sprint as A and D.

In one sentence: build a debt-and-spending risk monitor first (Clusters A + D combined), and treat credit education (Cluster B) as the natural next expansion once the core monitoring feature is live.